Tutorial 6
Torsional Members

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1. A stepped shaft $ABCD$ consisting of solid circular segments is subjected to three torques, as shown in the Fig. 1. The torques have magnitudes 12.5 k-in., 9.8 k-in., and 9.2 k-in. The length of each segment is 25 in. and the diameters of the segments are 3.5 in., 2.75 in., and 2.5 in. The material is steel with shear modulus of elasticity $G = 11.6 \times 10^3$ ksi.

   (a) Calculate the maximum shear stress τ_{max} in the shaft.

   (b) Calculate the angle of twist ϕ_D (in degrees) at end D.

   (c) Solve part (b) by using Energy Method

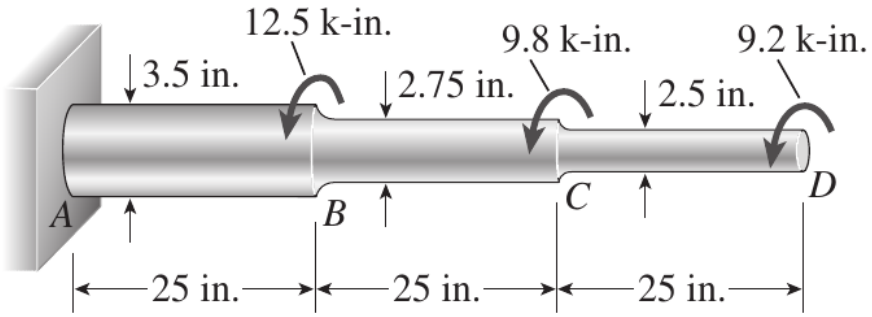


Fig. 1

2. A tapered bar AB of solid circular cross section is twisted by torques T (see Fig. 2). The diameter of the bar varies linearly from d_A at the left-hand end to d_B at the right-hand end.

   For what ratio d_B/d_A will the angle of twist of the tapered bar be one-half the angle of twist of a prismatic bar of diameter d_A? (The prismatic bar is made of the same material, has the same length, and is subjected to the same torque as the tapered bar.)

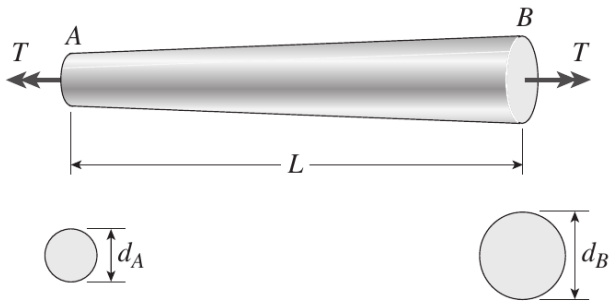


Fig. 2

3. The hollow shaft ① and the solid shaft ② are joint by a bolt at A as shown in Fig. 3. Determine the torque M_T and the twist angle β of the bolt after assembly for the case that the ends of the shafts have an angular difference of α in the stress-free state.

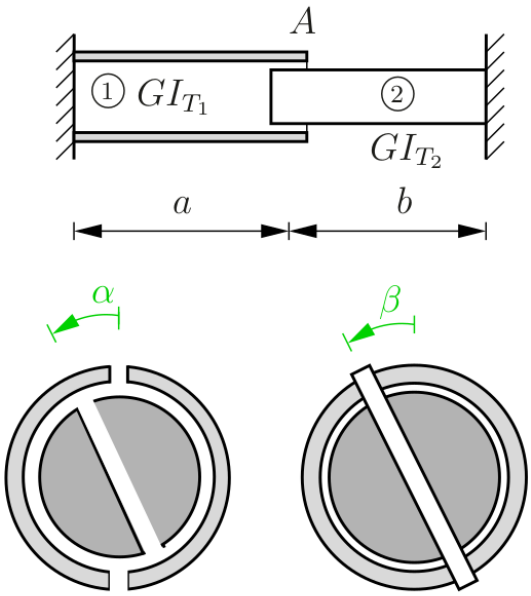


Fig. 3

4. A prismatic bar AB of solid circular cross section (diameter d) is loaded by a distributed torque (see Fig. 4). The intensity of the torque, that is, the torque per unit distance, is denoted $t(x)$ and varies linearly from a maximum value t_A at end A to zero at end B. Also, the length of the bar is L and the shear modulus of elasticity of the material is G.

   (a) Determine the maximum shear stress τ_{max} in the bar.

   (b) Determine the angle of twist ϕ between the ends of the bar.

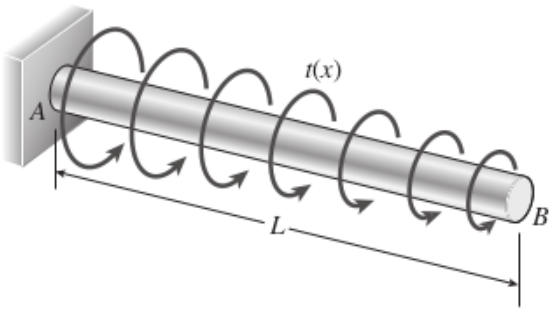


Fig. 4

5. A tube ② is mounted by heat shrinking on a shaft ① with circular cross section of different material. Determine the maximum shear stresses in ① and ② as well as the twist under the application of a torque M_T. (see Fig. 5)

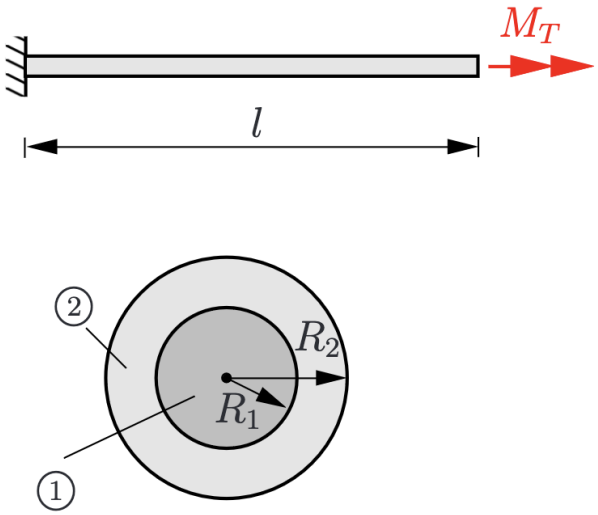


Fig. 5

6. The depicted gear-system consists of two shafts (lengths l_1, l_2) of identical material, that are connected by two gear wheels (radii R_1, R_2). The shaft ① (see Fig. 6) is loaded by an external torque M_1.

   (a) Determine M_2 such that equilibrium is fulfilled.

   (b) Choose the diameters d_1 and d_2 such that the admissible shear stress τ_{admis} is not exceeded?

   (c) Compute the angle of twist at position C, if shaft ② is fixed at position A.

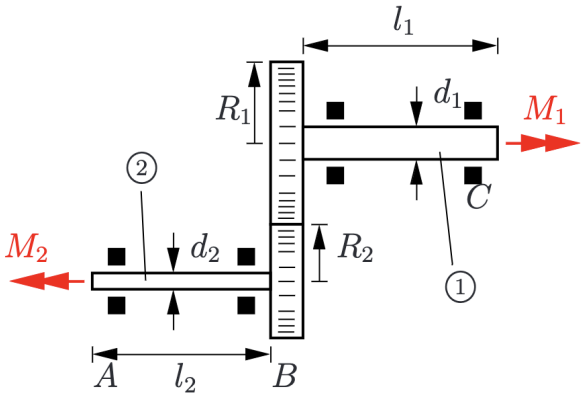


Fig. 6